Brownian Motion

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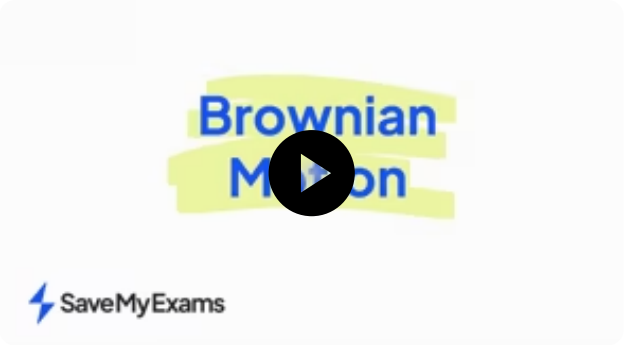

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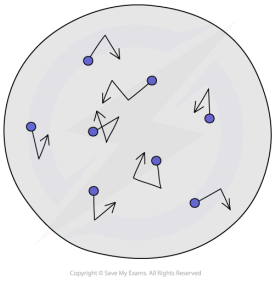
Help me understand

What is Brownian Motion?

- The **Kinetic Theory of Matter**, which simply says that **all matter is made up of tiny particles**, was discovered almost by accident
  - The Scottish scientist Robert Brown first described the random motion of pollen grains in water, which he saw under a microscope
  - This observation could not be explained at the time, but later it was realised that it shows that substances are made of particles which are in **constant motion** (hence 'kinetic')



Brownian Motion: the random motion of microscopic particles when observed through a microscope

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

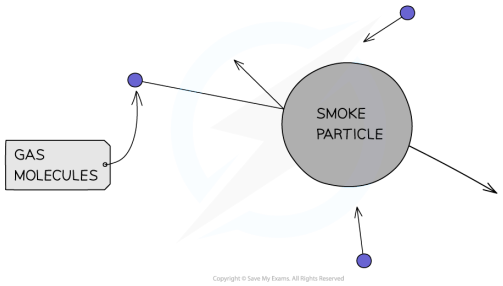
- When small particles (such as pollen or smoke) are suspended in a liquid or gas, they can be observed through a microscope moving around in a random, erratic fashion

Explaining Brownian motion

Extended tier only

What is Brownian motion caused by?

- When observing Brownian Motion, even with a microscope, only the **microscopic particles** can be seen
  - The pollen or smoke particles are seen to move
  - Smaller **atoms and molecules**, of water or air, are still too small to be seen
- These **light, fast-moving atoms and molecules** collide with the larger microscopic particles
  - The collisions give the particles a little nudge, causing them to change their speed and directions randomly, each time they are struck by a molecule
- The presence of the light, fast moving atoms and molecules is **inferred** from the motion of the microscopic particles
  - Inferences such as this are an important part of scientific investigation



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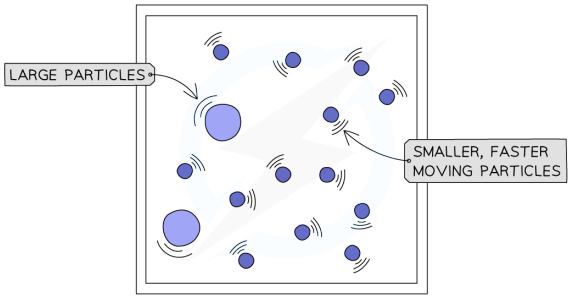
Light, fast-moving molecules collide with larger particles, giving them a little nudge



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